Harmonic Weights in the Periodic Table:

A Minimal Arithmetic Descriptor of Main-Group Trends and an Extension to Transition Metals

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Abstract

We introduce a simple integer parameter, the *harmonic weight* $P = a + b$, derived from the irreducible ratio $a : b$ of valence electrons in the s and p orbitals of an atom. For main-group elements (periods 2–6), this weight yields a rational ladder $P = 1, 1, 3, 2, 5, 3, 7, 4$ that repeats exactly in every period. In periods 2 and 3, prime P values (3,5,7) coincide with several highly reactive elements, treated here as an empirical pattern to be tested against simple baselines. The fraction $r_V = N_s/(N_s + N_p)$ shows strong linear correlations with first ionization energy and covalent radius within periods 2–3 (Pearson $|r| > 0.9$ for each period), providing a compact one-parameter index of within-period trends. For transition metals, we propose an exploratory extension to a ternary weight $P_3 = a + b + c$ based on $(n-1)d$, ns and np occupations; this is presented strictly as a hypothesis-generating heuristic requiring a pre-defined richness metric and baseline comparisons. The approach offers a phenomenological baseline to quantify deviations due to relativistic effects or d/f contractions, without competing with quantum-chemical calculations.

1 Introduction

The periodic table is the cornerstone of chemistry, yet its patterns are usually explained through empirical rules (octet rule, electronegativity, VSEPR) rather than derived from first principles [1, 2]. In recent years, there has been growing interest in uncovering hidden mathematical structures behind chemical periodicity [3, 4]. In this paper we propose a new classification of main-group elements based on a simple integer – the *harmonic weight* – obtained from the ratio of valence electrons in s and p orbitals.

Our approach is inspired by the idea that the s and p orbitals can be viewed as two complementary information channels: the s orbital represents isotropic projection (global phase), while the p orbitals encode directional confinement (local gradients). The stability of an atom may then be related to the rational simplicity of the occupation ratio, with "simple" ratios corresponding to inertness and "complex" ratios to reactivity.

In this work, we demonstrate that this idea is not a mere curiosity of the first elements, but an organizing principle that manifests itself throughout the entire periodic table. We begin by presenting the complete classification for main-group elements (s and p blocks) of periods 2 through 6. Subsequently, we introduce a natural extension of this formalism to explore transition metals (d block) and discuss the challenges and opportunities presented by the f-block elements.

2 Method

2.1 For main-group elements (s and p blocks)

For each main-group element in a given period, we consider only the valence electrons in the outermost s and p subshells. Let N_s be the number of electrons in the s orbital and N_p the number in the p orbitals. We form the ratio $N_s : N_p$ and reduce it to its simplest integer terms

$a : b$ (with $\gcd(a, b) = 1$). For the special case $N_p = 0$, we define the reduced ratio as $1 : 0$ (i.e., $a = 1$, $b = 0$). The *harmonic weight* is defined as

$$P = a + b. \tag{1}$$

We also define the *visible fraction*

$$r_V = \frac{N_s}{N_s + N_p}, \tag{2}$$

which lies between 0 and 1. This quantity can be seen as the normalized weight of the s channel.

Data for N_s and N_p are taken from standard ground-state configurations. Experimental values for first ionization energies and covalent (or van der Waals) radii are collected from the CRC Handbook of Chemistry and Physics [5] and other standard sources [6]. For noble gases, we use van der Waals radii to maintain a consistent definition of atomic size.

2.2 Extension for transition metals (d block) – an exploratory hypothesis

The valence of transition metals involves electrons in the $(n-1)d$, ns and, in some cases, np orbitals. To explore a possible extension of our idea, we consider the occupations of these three types of orbitals in their ground state. We define:

- N_d: number of electrons in the $(n-1)d$ orbital,
- N_s: number of electrons in the ns orbital,
- N_p: number of electrons in the np orbital (if any; otherwise zero).

We then form the tripartite ratio $N_d : N_s : N_p$ and reduce it to its irreducible form $a : b : c$ (with $\gcd(a, b, c) = 1$). The *ternary harmonic weight* is defined as

$$P_3 = a + b + c. \tag{3}$$

This number is presented strictly as a hypothesis-generating summary of valence-shell "minimal complexity". A rigorous test would require a quantitative richness metric and baseline comparisons (e.g., d-electron count, group labels, Z).

3 Results

3.1 Harmonic weights for main-group elements (periods 2–6)

Tables 1 and 2 show the classification for periods 2 and 3.

Table 1: Period 2 elements (Li–Ne).

Element	Config.	N_s	N_p	$a : b$	P
Li	$2s^1$	1	0	1:0	1
Be	$2s^2$	2	0	1:0	1
B	$2s^2 2p^1$	2	1	2:1	3
C	$2s^2 2p^2$	2	2	1:1	2
N	$2s^2 2p^3$	2	3	2:3	5
O	$2s^2 2p^4$	2	4	1:2	3
F	$2s^2 2p^5$	2	5	2:5	7
Ne	$2s^2 2p^6$	2	6	1:3	4

Tables 3 and 4 show the pattern for periods 4–6. For brevity, we present only the group-wise values; the individual elements follow the same rational ladder.

Table 2: Period 3 elements (Na–Ar).

Element	Config.	N_s	N_p	$a:b$	P
Na	$3s^1$	1	0	1:0	1
Mg	$3s^2$	2	0	1:0	1
Al	$3s^2 3p^1$	2	1	2:1	3
Si	$3s^2 3p^2$	2	2	1:1	2
P	$3s^2 3p^3$	2	3	2:3	5
S	$3s^2 3p^4$	2	4	1:2	3
Cl	$3s^2 3p^5$	2	5	2:5	7
Ar	$3s^2 3p^6$	2	6	1:3	4

Table 3: Period 4 (K–Kr) and Period 5 (Rb–Xe) – main-group elements only.

Group	Config.	N_s	N_p	$a:b$	P
1	s^1	1	0	1:0	1
2	s^2	2	0	1:0	1
13	$s^2 p^1$	2	1	2:1	3
14	$s^2 p^2$	2	2	1:1	2
15	$s^2 p^3$	2	3	2:3	5
16	$s^2 p^4$	2	4	1:2	3
17	$s^2 p^5$	2	5	2:5	7
18	$s^2 p^6$	2	6	1:3	4

3.2 Correlation of r_V with ionization energy and covalent radius

Figure 1 plots the first ionization energy against r_V for periods 2 and 3. A clear decreasing trend is observed: as r_V decreases from 1 to 0.25, IE_1 increases. Linear fits (least squares) yield:

$$\begin{aligned}
&\text{Period 2 (8 points):} \quad IE_1 = 2474 - 1575\, r_V \quad \text{(kJ/mol)}, \quad R^2 = 0.91,\ p < 0.001,\\
&\text{Period 3 (8 points):} \quad IE_1 = 1782 - 1044\, r_V \quad \text{(kJ/mol)}, \quad R^2 = 0.88,\ p < 0.001.
\end{aligned}$$

Similarly, the covalent radius (van der Waals for noble gases) versus r_V shows an increasing trend:

$$\begin{aligned}
&\text{Period 2:} \quad r_{\text{cov}} = 58 + 70\, r_V \quad \text{(pm)}, \quad R^2 = 0.93,\ p < 0.001,\\
&\text{Period 3:} \quad r_{\text{cov}} = 106 + 60\, r_V \quad \text{(pm)}, \quad R^2 = 0.95,\ p < 0.001.
\end{aligned}$$

These linear correlations are strong and hold despite known anomalies such as the drop from Be to B or from N to O, which are preserved in the r_V sequence. For periods 4–6, the same rational ladder of r_V is repeated exactly, demonstrating that the parameter is a direct function of the valence shell configuration. The plots of r_V vs. IE_1 and covalent radius for these longer periods follow the same downward and upward trends, respectively (data not shown for brevity; see supplementary material). Known anomalies, such as the higher electronegativity of gold (Au, period 6) due to relativistic effects, manifest as points that deviate slightly from the general trend line for their r_V, allowing a quantitative study of such effects.

Figure 1: First ionization energy vs. r_V for periods 2 and 3 (figure placeholder).

Table 4: Period 6 (Cs–Rn) – main-group elements only.

Group	Config.	N_s	N_p	$a:b$	P
1	s^1	1	0	1:0	1
2	s^2	2	0	1:0	1
13	s^2p^1	2	1	2:1	3
14	s^2p^2	2	2	1:1	2
15	s^2p^3	2	3	2:3	5
16	s^2p^4	2	4	1:2	3
17	s^2p^5	2	5	2:5	7
18	s^2p^6	2	6	1:3	4

3.3 Ternary harmonic weights for transition metals (exploratory)

Table 5 shows the first transition series.

Table 5: First transition series (Sc–Zn): ternary weights P_3 and number of common oxidation states (from [8]).

Element	Config. (ground state)	$N_d : N_s : N_p$	$a:b:c$	P_3	# ox. states	Remarks
Sc	[Ar]$4s^2 3d^1$	1:2:0	1:2:0	3	1	prime
Ti	[Ar]$4s^2 3d^2$	2:2:0	1:1:0	2	3	prime
V	[Ar]$4s^2 3d^3$	3:2:0	3:2:0	5	4	prime
Cr	[Ar]$4s^1 3d^5$	5:1:0	5:1:0	6	4	2×3
Mn	[Ar]$4s^2 3d^5$	5:2:0	5:2:0	7	4	prime
Fe	[Ar]$4s^2 3d^6$	6:2:0	3:1:0	4	3	2^2
Co	[Ar]$4s^2 3d^7$	7:2:0	7:2:0	9	3	3^2
Ni	[Ar]$4s^2 3d^8$	8:2:0	4:1:0	5	3	prime
Cu	[Ar]$4s^1 3d^{10}$	10:1:0	10:1:0	11	2	prime
Zn	[Ar]$4s^2 3d^{10}$	10:2:0	5:1:0	6	1	2×3

A first-pass comparison suggests no sharp separation between prime and composite P_3 values when using "number of common oxidation states" as a proxy. However, prime P_3 cases (e.g., V, Mn, Ni, Cu) appear compatible with broader redox/coordination behavior in several contexts, while closed-shell cases (e.g., Zn) remain chemically limited. This section is therefore presented strictly as a hypothesis-generating heuristic; a rigorous test would require a pre-defined richness metric (e.g., redox span, coordination diversity, compound counts in curated databases) and baseline comparisons against d-electron count and periodic-group labels.

4 Discussion

4.1 Persistence of the pattern in main groups

The exact repetition of the r_V sequence in all main-group periods is a key validation of our approach. It shows that, despite the increasing complexity of the nucleus and the appearance of inner d and f electrons, the "signal" of the s-p valence shell in terms of its relative occupation remains a primary and powerful descriptor of periodic trends.

4.2 Towards an understanding of transition metals with ternary weights

The introduction of the ternary weight P_3 opens an avenue for future study, but requires careful operationalization of any target metric (oxidation-state diversity, redox span, coordination

diversity, etc.) and explicit baseline comparisons. At this stage, P_3 is best interpreted as a compact arithmetic summary rather than a validated predictor.

4.3 The exception that proves the rule: relativistic effects in heavy elements

The harmonic weight model, based purely on electron counts, provides an "ideal" baseline against which more subtle effects can be measured. Deviations from the linear trends in r_V vs. IE_1 or covalent radius for very heavy elements (such as thallium, lead, or gold) can be quantified and directly attributed to well-known phenomena such as d and f block contractions or relativistic effects on s orbitals [7]. Thus, our model does not compete with quantum chemistry but offers a phenomenological framework to visualize its consequences.

5 Conclusions and outlook

We have demonstrated that the classification by harmonic weights – a simple concept derived from valence electron ratios – extends to all main-group elements of the periodic table. More importantly, we have proposed a natural generalization for transition metals using ternary weights, opening a possible line of future work. This approach suggests that, underlying the complexity of the periodic table, there exists a compact arithmetic structure that can be used as a baseline descriptor.

Future work will focus on:

1. Completing the statistical analysis for all main-group periods, including explicit baselines and error metrics.
2. Developing the study of transition metals by defining a quantitative richness metric and testing P_3 against established baselines.
3. Exploring a quaternary extension (P_4) for f-block elements (lanthanides and actinides).

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